Counting positive rational friezes over $\frac{1}{N}\mathbb{Z}$

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Abstract

Let $T(N,n)$ denote the number of positive rational frieze patterns of width n all of whose entries lie in $\frac{1}{N}\mathbb{Z}$. Conway and Coxeter showed $T(1,n) = C_{n-2}$. We prove $T(N,4) = d(2N^2)$, give closed counts at widths 5 and 6 together with proved search bounds, so that the tabulated values are complete rather than observed to stabilise, and tabulate $T(N,n)$ for widths 3 to 6, and for widths 8 and 9 at small N.

The structural result holds at every width. Homogenising the frieze continuant and applying the Desnanot–Jacobi identity collapses the lattice conditions to $gUV = N^{n-4}(U + V) + M$ with $M \mid N^2R$ and $N^{n-4}R \mid gM$, where R is a continuant window empty at width 5. The glide symmetry $m_{r,j} = m_{n-r,j+r}$ leaves only the rows $2 \le r \le \lfloor n/2 \rfloor$ independent. That the cutoff cannot be lowered is a finite check rather than a theorem.

At width 5 the reduction is the equation $xyz = A(x + y) + B$ of a problem posed by Ford generalising a question of Conrey, so $T(N,5)$ is governed by a named open problem in analytic number theory and the frieze count supplies a family of instances of it. We prove $T(N,5) = O_\varepsilon(N^{1+\varepsilon})$, and at primes $T(p,5) = O_\varepsilon(p^{1/3+\varepsilon})$ by importing the method of Conrey and Shah, together with $T(p,5) \ge 5\,d(p+1)$, which refutes the guess $T(N,5) \asymp d(N^2)\log^2 N$. Two obstructions to closing the gap are recorded: no asymptotic formula in $d(p+1)$ and powers of $\log p$ can hold, and the count is a count on the modular hyperbola $de \equiv 1 \bmod p$ whose governing Weil error exceeds the target.

The rotation action of $\mathbb{Z}/5$ on width-5 quiddity cycles is free, so $5 \mid T(N,5)$. At a prime the count splits as $T(p,5) = A(p) + B(p)$ with $3A(p) - 2B(p) = 15$ identically, and resolving the split gives $T(p,5) = 5 + 5C(p)$ for $p \ge 5$, where $C(p)$ counts the positive solutions of $auv = p + u + v$. Most of the paper is formalised in Lean 4, the count at a prime from the definition of a frieze itself; Section 9 says what is verified and what is not.

1 Introduction

A frieze pattern of width n is a finite array of rationals bordered by zeros and ones in which every adjacent 2×2 minor equals 1. Conway and Coxeter [4] classified the patterns with positive integer entries: they correspond to triangulations of a convex n-gon, so there are C_{n-2} of them. Allowing denominators changes the problem. Fixing N and asking for the positive patterns with all entries in $\frac{1}{N}\mathbb{Z}$ gives a count $T(N,n)$, finite because $\frac{1}{N}\mathbb{Z}$ is discrete and Cuntz and Holm [3] showed a discrete subset admits only finitely many friezes of each height, and computing it is the subject of this paper.

The point of this paper is that this count is an open problem in analytic number theory wearing different clothes. At a prime,

$$T(p,5) = 5 + 5C(p), \qquad C(p) = \#\{(a,u,v) \in \mathbb{Z}^3_{>0} : auv = p + u + v\}$$

(Theorem 7.7). The width-5 reduction is $guv = N(u+v) + M$, which is $xyz = A(x+y) + B$ with $A = N$ and $B = M$, posed by Ford in generalisation of a question of Conrey and recorded in [10]; the cubic $auv = p+u+v$ above is its case $A = 1$, $B = p$. Ford appears to have published nothing further on it. The fully symmetric variant $n = xyz + x + y + z$, also asked by Conrey, is Problem 25 of the American Institute of Mathematics problem list of May 2026 and is the equation of Conrey and Shah [8]. The order of $C(p)$ is not known. Counting friezes over $\frac{1}{p}\mathbb{Z}$ and estimating $C(p)$ are the same problem.

Friezes have been related to equations of this shape before, and the difference matters. Zhang [1] relates friezes to $xyz = G(x,y)$ in a different setting and direction: positive *integral* friezes of Dynkin type, where the Fontaine–Plamondon counts are determined exactly, and where the Diophantine conclusion is the existence of infinitely many positive solutions rather than a bound on their number. The equations here are the rational ones over $\frac{1}{N}\mathbb{Z}$, and the question is the counting bound asked by Conrey and, in generalised form, by Ford. Conley and Ovsienko [2] enumerate the positive *integer* solutions of the same matrix equation, graded by a parameter measuring the departure of a dissection from a triangulation; that parameter and the denominator N are independent, and none of the sequences below occurs in their tables.

What follows sets the limit of what that identification buys, and the limit is severe. The count is $O_\varepsilon(p^{1/3+\varepsilon})$ by the method of Conrey and Shah [8] (Theorem 8.3), while the conjectured order is $p^{o(1)}$ (Conjecture 8.4). Sections 8 onwards record what obstructs closing that gap. None of it is new to analytic number theory: the lower bounds of Propositions 8.5 and 8.6, $T(p,5) \geq 5\,d(p+1)$ and $C(p) \geq d(p+1) + d(2p+1)$, are the $a = 1$ and $a = 2$ slices, and Conrey and Shah record the same two slices for their equation in [8, §2.1], calling them easy to see and remarking that the size of the resulting sum of divisor functions is unclear. They are reproduced here because they calibrate the frieze count, not because they are new. Proposition 8.7 puts the count on the modular hyperbola $de \equiv 1 \bmod p$, where the Weil bound that governs such counts carries an error of size $p^{1/2}$ against a quantity of size $p^{1/3}$. The square-root barrier for box counts on a modular hyperbola is standard; what is recorded here is only that this problem sits on the wrong side of it.

The reader this is addressed to is a frieze theorist, not an analytic number theorist. To the latter the identification supplies instances of an open problem and no leverage on it. To the former it explains an absence: $T(N,4) = d(2N^2)$ has a closed form, counts of friezes over $\mathbb{Z}/n\mathbb{Z}$ are multiplicative with explicit formulas, and no such formula has been found at width 5 over $\frac{1}{N}\mathbb{Z}$. The reason is that finding one would settle a question of Conrey. Karpenkov, Short, van Son and Zabolotskii [5] classified exactly these friezes; the count they left open is hard for a reason external to frieze theory.

Section 7 tabulates $T(N,n)$ for widths 3 to 6 and for widths 8 and 9 at small N. At width 4 the count has a closed form, $T(N,4) = d(2N^2)$ (Theorem 4.1). The reductions and search bounds behind the tables at the larger widths are routine and are carried out in the accompanying library rather than here.

Section 9 says which results are machine-checked and which rest on a written proof; the per-statement map to Lean declarations is maintained with the library. Tables and computed constants are regenerated by scripts in `code/`, named at the point of use; `code/paper_numbers.py` collects those that have no script of their own.

2 Setting

We use the conventions of Karpenkov, Short, van Son and Zabolotskii [5].

Definition 2.1. A *rational frieze of width* n is a map $F\colon \{(i,j) : 0 \le i-j \le n\} \to \mathbb{Q}$, written $m_{i,j}$, with $m_{i,j} = 0$ when $i - j \in \{0, n\}$, with $m_{i,j} = 1$ when $i - j \in \{1, n-1\}$, and satisfying the diamond rule

$$m_{i,j}\, m_{i+1,j+1} - m_{i,j+1}\, m_{i+1,j} = 1.$$

It is *positive* if every entry with $0 < i-j < n$ is positive. The *quiddity cycle* is the row $i - j = 2$, a sequence $(a_0, \ldots, a_{n-1})$ read cyclically.

The quiddity cycle determines the frieze: writing r for the row index,

$$m_{r,j} = \frac{m_{r-1,j}\, m_{r-1,j+1} - 1}{m_{r-2,j+1}}, \tag{1}$$

with the closing conditions that row $n-1$ is constant 1 and row n is constant 0.

Definition 2.2. For $N \ge 1$ let $T(N, n)$ be the number of positive rational friezes of width n all of whose entries lie in $\frac{1}{N}\mathbb{Z}$. Quiddity cycles are counted as sequences, not up to rotation.

Conway and Coxeter [4] showed that the positive *integer* friezes of width n correspond to triangulations of a convex n-gon, so

$$T(1, n) = C_{n-2} = \frac{1}{n-1}\binom{2n-4}{n-2}. \tag{2}$$

Finiteness of $T(N, n)$ is Corollary 3.8 of Cuntz and Holm [3], which gives it for any discrete $R \subseteq \mathbb{C}$. Theorem B of [5] classifies the positive rational friezes over $\frac{1}{N}\mathbb{Z}$ in terms of minimal clockwise paths in a Farey-type graph. It classifies exactly the objects counted here, and does not count them. The present note is concerned with computing it.

3 The continuant parameterisation

Let K_k denote the frieze continuant, $K_k = x_k K_{k-1} - K_{k-2}$ with $K_0 = 1$, $K_{-1} = 0$. Continuants and the continued-fraction geometry behind them are treated in [6].

Proposition 3.1. *A positive rational frieze of width $n \ge 4$ is determined by $a_0, \ldots, a_{n-4}$, and*

$$a_{n-3} = \frac{K_{n-4}(a_0, \ldots, a_{n-5}) + 1}{K_{n-3}(a_0, \ldots, a_{n-4})}, \qquad a_{n-2} = K_{n-3}(a_0, \ldots, a_{n-4}),$$
$$a_{n-1} = \frac{K_{n-4}(a_1, \ldots, a_{n-4}) + 1}{K_{n-3}(a_0, \ldots, a_{n-4})}.$$

Proof. The closing conditions are $m_{n-1,j} = 1$ and $m_{n,j} = 0$ for every j, that is $K_{n-2}(a_j, \ldots, a_{j+n-3}) = 1$ and $K_{n-1}(a_j, \ldots, a_{j+n-2}) = 0$. Expanding the second at $j = 0$ by the recursion $K_k = x_k K_{k-1} - K_{k-2}$ in its last argument and solving for a_{n-2} gives the middle display; expanding the first at $j = 0$ and at $j = 1$ and solving for a_{n-3} and a_{n-1} gives the other two. The remaining $n-3$ closing conditions are cyclic conjugates of these under the monodromy $M(a_0) \cdots M(a_{n-1}) = -I$ and impose nothing further. □

At $n = 5$ this reads

$$a_2 = \frac{a_0 + 1}{D}, \qquad a_3 = D, \qquad a_4 = \frac{a_1 + 1}{D}, \qquad D = a_0 a_1 - 1, \tag{3}$$

and at $n = 6$, with $D = K_3(a_0, a_1, a_2) = a_2(a_0 a_1 - 1) - a_0$,

$$a_3 = \frac{a_0 a_1}{D}, \qquad a_4 = D, \qquad a_5 = \frac{a_1 a_2}{D}. \tag{4}$$

Remark 3.2. A frieze of width n has a glide symmetry exchanging row r with row $n - r$. For $n = 5$ the only non-constant interior rows are 2 and 3, and they are exchanged, so a quiddity cycle in $\frac{1}{N}\mathbb{Z}$ forces every entry into $\frac{1}{N}\mathbb{Z}$. For $n = 6$ row 3 is fixed by the glide, and carries a lattice condition of its own. This is the source of the extra divisibility constraints at width 6.

4 Width four

Write d for the number-of-divisors function.

Theorem 4.1. *For every $N \geq 1$, $T(N, 4) = d(2N^2)$.*

Proof. Let (a_0, a_1, a_2, a_3) be the quiddity cycle. By (1) the row 3 entries are $a_j a_{j+1} - 1$, and row 3 is constant 1, so $a_j a_{j+1} = 2$ for all j modulo 4. Hence $a_0 = a_2$, $a_1 = a_3$ and $a_0 a_1 = 2$. Conversely any $a > 0$ with $a \cdot (2/a) = 2$ yields the frieze with quiddity $(a, 2/a, a, 2/a)$, whose remaining rows are forced and positive.

Write $a_0 = p/N$ with p a positive integer. Then $a_1 = 2N/p$, and $a_1 \in \frac{1}{N}\mathbb{Z}$ if and only if $2N^2/p \in \mathbb{Z}$, that is $p \mid 2N^2$. Distinct divisors give distinct cycles, so $T(N, 4)$ is the number of positive divisors of $2N^2$. □

The sequence $d(2N^2)$ is OEIS A361689.

5 Width five

Put $a_0 = p/N$ and $a_1 = q/N$ with p, q positive integers, and set $e := pq - N^2$.

Proposition 5.1. *The parameters (p, q) arise from a positive frieze of width 5 over $\frac{1}{N}\mathbb{Z}$ if and only if*

$$e > 0, \qquad N \mid e, \qquad e \mid N^2(p + N), \qquad e \mid N^2(q + N). \tag{5}$$

Proof. By (3), $a_3 = D = e/N^2$, $a_2 = N(p + N)/e$ and $a_4 = N(q + N)/e$. Positivity of the frieze is equivalent to $D > 0$, that is $e > 0$. Membership of a_3 in $\frac{1}{N}\mathbb{Z}$ is $N \mid e$, and membership of a_2 and a_4 is the two divisibility conditions. By Remark 3.2 no further condition is needed. □

Lemma 5.2 (Search bound)**.** *Any solution of* (5) *satisfies $p \leq N^3 + 2N^2$ and $q \leq N^3 + 2N^2$.*

Proof. Since $e > 0$ and $e \mid N^2(q+N)$ with $N^2(q+N) > 0$, we have $e \leq N^2(q+N)$. Substituting $e = pq - N^2$ gives $pq - N^2 \leq N^2 q + N^3$, that is $q(p - N^2) \leq N^3 + N^2$. As $q \geq 1$, either $p \leq N^2$ or $p - N^2 \leq N^3 + N^2$; in both cases $p \leq N^3 + 2N^2$. The bound on q is symmetric. □

Lemma 5.2 makes the enumeration finite with an explicit range. Since additionally $e \mid N^2(p + N)$ and $q = (e + N^2)/p$, one may loop over p in the proved range and over the divisors e of $N^2(p + N)$, which is what `code/verify_width5_proved.py` does. The resulting values of $T(N, 5)$ for $N \leq 40$ are recorded in `data/T5.txt`. They are complete, not merely stable under an increasing cutoff.

6 Symmetry at width five

The dihedral group D_5 acts on width-5 quiddity cycles by rotation and reversal.

Theorem 6.1. *The rotation action of $\mathbb{Z}/5$ on positive rational width-5 quiddity cycles is free. In particular $5 \mid T(N,5)$ for every N.*

Proof. The closing condition at width 5 is $(a_j a_{j+1} - 1)(a_{j+1}a_{j+2} - 1) = 1 + a_{j+1}$ for all j modulo 5. A cycle fixed by a nontrivial rotation is constant, say $a_j = a$, whence $(a^2-1)^2 = 1 + a$, that is $a(a^3 - 2a - 1) = 0$. Since $a > 0$ and $a^3 - 2a - 1 = (a+1)(a^2 - a - 1)$, the only positive root is the golden ratio $(1+\sqrt{5})/2$, which is irrational. So no positive rational cycle is fixed. $\square$

7 Data and questions

Every value below is produced by a procedure whose search range is proved, so the counts are complete rather than stable under an increasing cutoff.

	$n=3$	$n=4$	$n=5$	$n=6$
$N=1$	1	2	5	14
$N=2$	1	4	20	102
$N=3$	1	6	40	259
$N=4$	1	6	60	487

Neither $T(N,5)$ nor $T(N,6)$ appears in the OEIS as a function of N, and neither is multiplicative: $T(2,5)T(3,5) = 800 \neq 110 = T(6,5)$.

At widths 8 and 9 the same reduction, applied with the glide symmetry as a pruning rule and swept over the entry box of Cuntz and Holm [3], gives

N	1	2	3	4
$T(N,8)$	132	2576	9980	28604
$T(N,9)$	429	13101		

(`code/w89_count.rs`). The values at $N=1$ are C_6 and C_7, as they must be, and neither sequence appears in the OEIS. The sweep is complete over a proved box, so these are counts rather than stable observations; the cost grows like $N^{3(n-3)}$, which is why the table stops where it does. The reduction and the search bound behind it are routine and are carried out in the accompanying library rather than here.

Theorem 7.1 (Reduced form)**.** *For every $N \geq 1$,*

$$T(N,5) = \#\{(u,v,M) \in \mathbb{Z}^3_{>0} \;:\; \gcd(u,v) = 1,\; M \mid N^2,\; uv \mid N(u+v) + M,\; Nuv \mid M\big(N(u+v)+M\big)\}.$$

Proof. Multiplying $gu > N$ by $v > 0$ gives $guv > Nv$, that is $Nu + M > 0$, which holds for $u, M > 0$; the same argument disposes of $gv > N$. So both positivity conditions are automatic. For the equation, g is determined by (u,v,M), so requiring some g with $guv = N(u+v) + M$ is requiring $uv \mid N(u+v)+M$. Given that, multiplying $N \mid gM$ by uv gives $Nuv \mid M(N(u+v)+M)$, and the two are equivalent because $uv \neq 0$. $\square$

Proposition 7.2 (Hyperbola bound)**.** *Every solution counted by Theorem 7.1 satisfies* $(u-N)(v-N) \le N^2 + M \le 2N^2$*, and if* $u \le v$ *then* $u < 3N$*.*

Proof. Since uv divides the positive integer $N(u+v)+M$ it is at most that integer, and $(u-N)(v-N) = uv - N(u+v) + N^2$, so $(u-N)(v-N) \le N^2 + M$; and $M \le N^2$ because $M \mid N^2$. If $u \le N$ then $u < 3N$. Otherwise $0 < u - N \le v - N$, so $(u-N)^2 \le (u-N)(v-N) \le 2N^2$, whereas $u \ge 3N$ would force $(u-N)^2 \ge 4N^2$. □

The change is one of order: the smaller parameter is bounded *linearly* in N, where Lemma 5.2 gives only $N^3 + 2N^2$ for the original parameters. The bound $(u-N)(v-N) \le 2N^2$ is attained for every $2 \le N \le 24$ (`code/t5_bounds.py`).

Corollary 7.3. $T(N,5) \;\le\; 2 \sum_{M \mid N^2} \sum_{1 \le u < 3N} d(Nu+M)$*, and consequently* $T(N,5) = O_\varepsilon\big(N^{1+\varepsilon}\big)$*.*

Proof. By Proposition 7.2 the smaller of u, v is less than $3N$; fix it as u and let v run over the divisors of $Nu + M$, which is legitimate because $v \mid N(u+v)+M$ and $v \mid Nv$ together give $v \mid Nu + M$. Each unordered pair is counted at most twice. For the second statement, $d(n) = O_\varepsilon(n^\varepsilon)$ gives $d(N^2) = O_\varepsilon(N^\varepsilon)$ and $d(Nu+M) = O_\varepsilon(N^\varepsilon)$ uniformly for $u < 3N$ and $M \le N^2$, so the double sum is $O_\varepsilon(N^{1+\varepsilon})$. □

No matching lower bound of the same order appears to exist.

Definition 7.4. Let p be prime. In the parametrisation of Theorem 7.1 the parameter M divides p^2, so $M \in \{1, p, p^2\}$, and $M = 1$ cannot occur: the condition $Nuv \mid M(N(u+v)+M)$ reads $p\,uv \mid p(u+v)+1$ there, forcing $p \mid 1$. Write $A(p)$ for the number of solutions with $M = p$ and $B(p)$ for the number with $M = p^2$, so that $T(p,5) = A(p) + B(p)$. At $M = p$ both divisibility conditions of Theorem 7.1 reduce to $uv \mid p(u+v+1)$, so

$$A(p) = \#\{(u,v) \in \mathbb{Z}_{>0}^2 : \gcd(u,v) = 1,\ uv \mid p(u+v+1)\}.$$

A quiddity cycle has $M = p$ exactly when its parameters do, so this is also the count of cycles with $M = p$ used in Theorem 7.7.

Proposition 7.5. *Let* $p, u, v \ge 1$ *with* $\gcd(uv,p) = 1$ *and* $uv \mid p(u+v+1)$*. Then*

$$(u,v) \in \{(1,1),\ (1,2),\ (2,1),\ (2,3),\ (3,2)\},$$

independently of p*. These five pairs form a single orbit of the rotation action of* $\mathbb{Z}/5$ *of Theorem 6.1; that last assertion is not proved here and nothing below depends on it.*

Proof. Since $u \mid uv \mid p(u+v+1)$ and $\gcd(u,p) = 1$ we get $u \mid u+v+1$, hence $u \mid v+1$; symmetrically $v \mid u+1$. So $u \le v+1$ and $v \le u+1$. If $u = v$ then $u \mid u+1$ gives $u = 1$. If $v = u+1$ then $u \mid v+1 = u+2$ gives $u \mid 2$, so (u,v) is $(1,2)$ or $(2,3)$; the case $u = v+1$ is symmetric. Coprimality of u and v is not needed. □

So every further solution at a prime requires $p \mid uv$, and $T(p,5)$ stays small: it is 160 at $p = 541$ and 130 at $p = 421$, against $T(120,5) = 4170$.

By Remark 3.2 the width-5 glide gives $a_{j+3} = a_j a_{j+1} - 1$ for every j modulo 5, which in numerators is

$$p_j p_{j+1} = p\,(p_{j+3} + p) \qquad (j \in \mathbb{Z}/5), \tag{6}$$

so $e_j = p\,p_{j+3}$ and therefore $M_j = p\,p_{j+3}/g_j$ with $g_j = \gcd(p_j + p,\, p_{j+1} + p)$. In particular $M_j = p^2$ forces $p \mid p_{j+3}$, so the M-profile is controlled by which entries of the quiddity are integers.

Proposition 7.6. *Let p be prime. The number of integer entries in the quiddity cycle of a positive width-5 frieze over $\frac{1}{p}\mathbb{Z}$ is either 3 or 5, and it is 5 exactly for the Conway–Coxeter frieze $(2, 2, 1, 3, 1)$ and its rotations.*

Proof. Put $I = \{j : p \mid p_j\}$. If $p \nmid p_{j+3}$ then the right side of (6) has p-valuation exactly one, so exactly one of p_j, p_{j+1} is divisible by p, and that one has valuation exactly one. Taking $j = m + 2$, this says: for every $m \notin I$, exactly one of $m + 2$, $m + 3$ lies in I. Exhausting the 32 subsets of $\mathbb{Z}/5$, the only ones with that property have cardinality 3 or 5, and those of cardinality 3 are exactly the sets $\{i, i + 1, i + 3\}$. If $|I| = 5$ every entry is an integer, so the frieze is a Conway–Coxeter frieze of width 5, and there are $C_3 = 5$ of those, forming one rotation orbit. □

Theorem 7.7 (Orbit split)**.** *Let p be prime and write $R = T(p, 5)/5$ for the number of rotation orbits. Each orbit contains exactly two cycles with $M = p$ and three with $M = p^2$, except the orbit of the Conway–Coxeter frieze $(2, 2, 1, 3, 1)$, where all five have $M = p$. Hence $A(p) = 2R+3$ and $B(p) = 3R - 3$, so that*

$$3A(p) - 2B(p) = 15 \qquad \textit{and} \qquad T(p, 5) = \tfrac{1}{2}\big(5A(p) - 15\big).$$

Proof. By Proposition 7.6, $I = \{j : p \mid p_j\}$ has three or five elements. If it has five the frieze is $(2, 2, 1, 3, 1)$ up to rotation, and evaluating M_j at each of the five positions gives p every time.

Suppose $|I| = 3$. The condition established in the proof of Proposition 7.6, that every $m \notin I$ has exactly one of $m+2$, $m+3$ in I, forces $I = \{i, i+1, i+3\}$ for some i. That proof also records that the divisible one of p_j, p_{j+1} has valuation exactly one whenever $p \nmid p_{j+3}$; applying this at $j = i-1$ and at $j = i+1$, where p_{i+2} respectively p_{i+4} is prime to p, gives $v_p(p_i) = v_p(p_{i+1}) = 1$ exactly, while $v_p(p_{i+2}) = v_p(p_{i+4}) = 0$ because $i + 2, i + 4 \notin I$. Now take the five positions in turn.

For $j = i + 1$ and $j = i + 4$ we have $j + 3 \notin I$. In each case one of the two entries p_j, p_{j+1} is prime to p, hence so is g_j, and $v_p(M_j) = 1 + 0 - 0 = 1$, so $M_j = p$.

For $j = i+2$ and $j = i+3$ the entry p_{i+2} respectively p_{i+4} is prime to p, so again g_j is prime to p, while $v_p(p_{j+3}) = 1$; thus $v_p(M_j) = 2$ and $M_j = p^2$.

For $j = i$ write $p_i = p\alpha$ and $p_{i+1} = p\beta$, so $g_i = pG$ with $G = \gcd(\alpha + 1, \beta + 1)$, and (6) at $j = i$ gives $p_{i+3} = p(\alpha\beta - 1)$ with $\alpha\beta - 1 > 0$. If $M_i = p$ then $g_i = p_{i+3}$, that is $G = \alpha\beta - 1$. Since $G \mid \alpha + 1$ this makes $\alpha\beta - 1$ divide $\alpha + 1$, say $\alpha + 1 = (\alpha\beta - 1)s$. But (6) at $j = i + 2$ reads $p_{i+2}p_{i+3} = p(p_i + p)$, that is $p_{i+2}(\alpha\beta - 1) = p(\alpha + 1)$, so $p_{i+2}(\alpha\beta - 1) = p(\alpha\beta - 1)s$, and cancelling $\alpha\beta - 1 \neq 0$ gives $p_{i+2} = ps$, contradicting $p \nmid p_{i+2}$. Hence $M_i = p^2$.

So two positions carry $M = p$ and three carry $M = p^2$. Counting over the R orbits, $A(p) = 2(R - 1) + 5$ and $B(p) = 3(R - 1)$, and $T(p, 5) = A + B = 5R$. □

8 The width-five count at a prime

No closed form for $T(p, 5)$ should be expected. Setting $u = ps$ in the $M = p$ term and writing $v + 1 = st$, the condition $uv \mid p(u + v + 1)$ becomes $st - 1 \mid p + t$, and putting $p + t = (st - 1)m$ turns it into the symmetric cubic

$$stm - t - m = p, \qquad \text{equivalently} \qquad (st - 1)(sm - 1) = sp + 1. \tag{7}$$

The correspondence is exact.

Lemma 8.1. *Let $p \geq 5$ be prime and let $C(p)$ be the number of triples (s, t, m) of positive integers with $stm = p + t + m$. Then $A(p) = 5 + 2C(p)$.*

Proof. By Definition 7.4, $A(p)$ counts the coprime pairs (u, v) with $uv \mid p(u+v+1)$, a condition symmetric in u and v. If $\gcd(uv, p) = 1$ it reduces to $uv \mid u + v + 1$, and Proposition 7.5 lists the five pairs satisfying it. Otherwise $p \mid uv$, and $\gcd(u, v) = 1$ forces p to divide exactly one of u and v; by the symmetry the two halves are equinumerous, so $A(p) - 5 = 2\,\#\{(u, v) : p \mid u\}$.

For the pairs with $p \mid u$ write $u = ps$. Then $uv \mid p(u + v + 1)$ reads $sv \mid ps + v + 1$, whence $s \mid v+1$; put $v = st-1$ with $t \geq 1$. The condition becomes $s(st-1) \mid s(p+t)$, that is $st-1 \mid p+t$; put $p + t = (st - 1)m$ with $m \geq 1$. Expanding, $stm = p + t + m$, which is (7).

Conversely, given (s, t, m) with $stm = p + t + m$, set $u = ps$ and $v = st - 1$. Then $st - 1 = (p + t)/m \geq 1$, so $v \geq 1$, and $\gcd(u, v) = 1$: indeed $\gcd(s, st - 1) = 1$, and $p \nmid st - 1$, since $st \equiv 1 \bmod p$ would give $m \equiv t + m$, hence $p \mid t$ and then $st \equiv 0$, a contradiction. The two maps are mutually inverse, so $\#\{(u, v) : p \mid u\} = C(p)$. □

Checked against $A(p)$ computed directly for the primes $5 \leq p \leq 53$ (`code/a_count.py`), with no discrepancy. Equation (7) also bounds the count, and does so better than Corollary 7.3.

The exponent can be lowered to $\frac{1}{3}$ by importing the method of Conrey and Shah [8], who bound the number of representations $n = xyz+x+y+z$ by $n^{1/3} \log n (\log\log n)^4$. Their argument uses that full symmetry forces the smallest variable below $n^{1/3}$. Our equation $suv = p + u + v$ is symmetric only in u and v, but the same bound holds.

Lemma 8.2. *For $p \geq 4$, every solution of $suv = p+u+v$ in positive integers has $\min(s, u, v)^3 \leq 2p$.*

Proof. If $\min(s, u, v)^3 > 2p$ then $suv > 2p$, so $p < u+v$ and hence $suv = p+u+v < 2(u+v) \leq 4\max(u, v)$. Cancelling the maximum leaves a product of two of the variables below 4. But $\min(s, u, v)^3 > 2p \geq 8$ forces every variable to be at least 3, so that product is at least 9. □

Theorem 8.3. $T(p, 5) = O_\varepsilon\big(p^{1/3+\varepsilon}\big)$.

Proof. By Lemma 8.2 some variable is at most $(2p)^{1/3}$. If it is $s = t$, then $(tu-1)(tv-1) = tp+1$ determines (u, v) from a divisor of $tp+1$. If it is $u = t$ or $v = t$, then $v(su-1) = p+u$ determines the pair from a divisor of $p + t$. Hence

$$C(p) \;\leq \sum_{1 \leq t \leq (2p)^{1/3}} \big(d(tp + 1) + 2\, d(p + t)\big),$$

and a divisor-sum bound of Shiu type [9] gives $\ll p^{1/3} \log p\, (\log\log p)^c$. The conclusion follows from $T(p, 5) = 5 + 5C(p)$ of Theorem 7.7. □

The proof of Theorem 8.3 loses its exponent at a single step, and that step can be isolated. Writing $C(p) = \sum_t \#\{w \mid p + t : t \mid w + 1\}$, each term is at most $d(p + t) = p^{o(1)}$, so

$$C(p) \;\leq\; \#\{t : \text{the } t\text{-th term is nonzero}\} \;\cdot\; \max_t d(p + t),$$

and Conjecture 8.4 at a prime is equivalent to the assertion that the first factor is $p^{o(1)}$. The proof bounds that factor by $p^{1/3}$; the truth is much smaller. Over the 6055 primes $5 \leq p < 6 \cdot 10^4$, taken in five blocks of equal size, the ratio of that count to $\log^2 p$ has block means 0.707, 0.669, 0.662, 0.663, 0.658, while its ratio to $p^{1/3}$ falls from 3.05 to 2.07 across the same range (`code/atom_b.rs`, `code/atom_b_stats.py`).

The count is far smaller than $p^{1/3}$, but only above the point where $p^{1/3}$ exceeds $\log^2 p$, near $p = 10^7$. Over eight primes between 10^9 and $2.4 \cdot 10^9$ it lies between 0.067 and 0.128 times $p^{1/3}$, while its ratio to $\log^2 p$ stays between 0.19 and 0.30 (`code/s2_witness.rs`). The natural elementary route to a saving does not work, and the reason is quantitative. Splitting the contributing t by the least k with $kt - 1 \mid p + t$, the terms with $k \le K$ are governed by divisors of $kp + 1$ and number $O(K \log p)$; the terms with $k > K$ have a complementary divisor $v < p/K$ and satisfy $t \mid p + v$, so they number $O((p/K) \log p)$. Balancing gives $K = \sqrt{p}$ and a total of $O(\sqrt{p} \log p)$, which is worse than the trivial bound $p^{1/3}$; the balance is `split_cost_ge`, which shows $4p \le (K + q)^2$ whenever $p \le Kq$, so no choice of K brings the two branches below $2\sqrt{p}$. At $p \approx 10^9$ the least such k reaches $7 \cdot 10^7$, so the second branch is not empty.

The count admits a description in terms of fractional parts. For a divisor D of $p + t$ the shift is forced, $t = \lceil p/D \rceil D - p = D(1 - \{p/D\})$, and the condition becomes $t \mid D + 1$. So the quantity to bound is the number of D for which $\lceil p/D \rceil D - p$ divides $D + 1$, exactly for $D \ge 2$. This is a statement about the distribution of $\{p/D\}$, and the pair (t, D) it produces satisfies $t \mid D+1$ and $D \mid p+t$, which is the system one started from: the elementary reindexings of this problem are fixed points, which is why they do not reduce it.

It is also worth recording that Corollary 7.3 is already optimal given that state of the art. Its per-M input is a bound of size $N \log N$ for the solutions of $guv = N(u + v) + M$, and with $|AB| = NM \le N^3$ that is $|AB|^{1/3}$ up to logarithms, exactly the strength of [8]. Improving Corollary 7.3 therefore requires progress on the problem Ford posed.

For all 123 primes $5 \le p < 700$ the ratio $T(p, 5)/\sqrt{p}$ has mean 15.2, and 12.0 over $p > 400$ (`code/t5_primes_split.py`); it does not decay on that range. Theorem 8.3 forces it to decay eventually, and the paragraph above places the onset near $p = 10^7$, so the small-prime data says nothing against it.

The conjecture is not new, and it is not ours. Conrey asked whether the number of positive integer solutions of $xyz + x + y = n$ is $O_\varepsilon(n^\varepsilon)$, and Ford generalised this to $xyz = A(x + y) + B$ with $O_\varepsilon(|AB|^\varepsilon)$ solutions; both questions are recorded in [10]. Our width-5 equation $guv = N(u + v) + M$ is Ford's equation with $A = N$ and $B = M$, so Conjecture 8.4 below is a case of Ford's problem, summed over $M \mid N^2$. Writing $R_a(n) = \#\{(x, y) \in \mathbb{N}^2 : axy - x - y = n\}$ for the quantity studied in [10], at a prime

$$C(p) = \sum_{a \ge 1} R_a(p), \qquad R_1(p) = d(p + 1),$$

both verified for every prime $p < 200$. The second identity is Proposition 8.5: our lower bound is the $a = 1$ term. Huang proves an asymptotic for $\sum_{n \le N} R_a(n)$ with a fixed, which is the averaged form of the inner sum.

That the frieze count is governed by a problem of Conrey and Ford is the reason Conjecture 8.4 is stated and not proved here.

Conjecture 8.4. $T(N, 5) = N^{o(1)}$.

The natural sharper guess, $T(N, 5) \asymp d(N^2) \log^2 N$, is false, and the reason is a lower bound that the numerics cannot see.

Proposition 8.5. *$C(p) \geq d(p+1)$, hence $T(p,5) \geq 5\,d(p+1)$, for every prime p. Consequently $T(N,5) \asymp d(N^2)\log^2 N$ fails.*

Proof. Taking $u = 1$ in (7), written as $suv = p + u + v$, leaves $sv = p + 1 + v$, which is solvable exactly when $v \mid p+1$, with $s = (p+1)/v + 1$. Distinct divisors give distinct solutions, so $C(p) \geq d(p+1)$, and $T(p,5) = 5 + 5C(p)$ by Theorem 7.7.

For the consequence, let m_r be the product of the first r primes, so $d(m_r) = 2^r$ and $\log m_r \sim r \log r$. By Linnik's theorem there is a prime $p \equiv -1 \pmod{m_r}$ with $\log p \ll \log m_r$. Then $m_r \mid p+1$, so $d(p+1) \geq 2^r$ while $\log^2 p \ll (r\log r)^2$, and $2^r/(r\log r)^2 \to \infty$. Since $d(p^2) = 3$ at a prime, the asserted equivalence would force $d(p+1) = O(\log^2 p)$. □

The crossover is far out of computational reach: 2^r overtakes $(r\log r)^2$ near $r = 20$, where m_r already exceeds 10^{25}. So the flat ratios observed for $N \leq 3000$ are what the proposition predicts, not evidence against it. This is why the equivalence form survived the out-of-sample test of the previous paragraph and is nonetheless false: numerical stability over three decades is not an asymptotic.

The heuristic is that for each $M \mid N^2$ and each u, the admissible v are the divisors of $Nu+M$ in a fixed residue class modulo u, of which there are $d(Nu+M)/u$ on average; summing $1/u$ over u and averaging d contributes one logarithm each.

The statement was frozen on the range $N \leq 600$ and then tested on $601 \leq N \leq 3000$, a disjoint range four times larger. The mean of $T(N,5)/\big(d(N^2)\log^2(N+2)\big)$ moves from 2.979 to 2.838, a shift of 0.953, and its block means over the new range are 2.847, 2.832, 2.835, 2.839. Every rival normalisation drifts on the same test: $d(N^2)\sqrt{N}$ by a factor 0.676, $d(N^2)\log N$ by 1.313, $d(N^2)\log^3 N$ by 0.657, and $T(N,5)/N$ by 0.424 (`code/t5_growth.rs`, `data/T5_growth.csv`). Over the full range $\min T(N,5)/N$ falls to 0.0673 at $N = 2971$, so $T(N,5) = o(N)$ is not in doubt. This is evidence, not a proof: the numerics are out of sample, and no proof is offered.

The obstruction can be made exact. Counting by (g, M) rather than (u, v) turns the width-5 count into a count of divisors in residue classes, with no frieze structure left in it.

The bound $g \leq N^2 + 2N$ is attained, and is exactly the largest g occurring, for every $6 \leq N \leq 25$. The correspondence has been checked for every $N \leq 25$: the refined count is $T(N,5)$ on the nose, and the upper bound obtained by dropping the gcd condition overshoots by a factor between 1.95 and 7.33 on that range (`code/paper_numbers.py`).

So Conjecture 8.4 is the statement that $\sum_{M\mid N^2}\sum_{g\leq N^2+2N} \#\{D \mid N^2 + gM : D \equiv -N \bmod g\}$, refined by the gcd condition, is $N^{o(1)}$. Every elementary route available bounds this by $O(N^{1+o(1)})$ and no further: pairing divisors gives $\#\{D\} \leq 2\big(\sqrt{N^2+gM}/g + 1\big)$, and the $+1$ alone sums to $\Theta(N)$. Removing it means showing that for most g no divisor of $N^2 + gM$ lies in the prescribed class, which is a statement about divisors in residue classes uniform in the modulus. It is worth recording that the per-M counts are *not* of size $N^{o(1)}$: at $N = 420$ a single M contributes 765 ordered pairs (`code/paper_numbers.py`), so any proof must exploit cancellation across M rather than bound each term separately. Multiplying $auv = p + u + v$ through by a gives

$$(au-1)(av-1) = ap + 1, \tag{8}$$

so the solutions with leading coefficient a are the divisors of the shifted number $ap+1$ lying in the class $-1 \bmod a$, and

$$C(p) = \sum_{a\geq 1} \#\{D \mid ap+1 \,:\, D \equiv -1 \bmod a\}. \tag{9}$$

The term $a = 1$ is $d(p+1)$, the same count as the $u = 1$ family of Proposition 8.5 though a different family. The term $a = 2$ is the one that costs nothing: $2p+1$ is odd, so every divisor of it is odd and hence of the form $2u - 1$, and no divisor is lost to the congruence. Thus $R_2(p) = d(2p+1)$ exactly, and since the two families carry different values of a they are disjoint.

Proposition 8.6. *$C(p) \geq d(p+1) + d(2p+1)$.*

Proof. Formula (8) at $a = 1$ and $a = 2$; the two families are disjoint because a is determined by the solution. Formalised as `cubic_lower_two_families`. □

The upper-bound half of (9) over a box is `cubic_shift_decomp`, with the factorisation (8) as `cube_factor_nat` and the injectivity of the fibre as `cube_det`. It is the decomposition every upper bound passes through: the a-th term is at most $d(ap+1)$, which the divisor bound of `card_divisors_pow_le` controls, and $\min(a, u, v)^3 \leq 2p$ caps the range of a at $(2p)^{1/3}$, giving $C(p) \ll_\varepsilon p^{1/3+\varepsilon}$.

The fibres carry a constraint that is group-theoretic rather than analytic. Reducing mod a, the divisors of $ap+1$ land in the multiplicative subsemigroup of $(\mathbb{Z}/a)^\times$ generated by the prime factors of $ap+1$, so the a-th fibre is empty unless -1 lies in that subsemigroup.

The full subsemigroup criterion was measured against the fibres directly, for a up to $(2p)^{1/3}$ and p sampled over four decades. It never once failed, and it accounts for 28.6%, 31.7%, 31.4%, 33.1% of all a at $p \approx 10^4, 10^5, 10^6, 10^7$. The proportion does not grow, so the criterion improves the implied constant in $C(p) \ll_\varepsilon p^{1/3+\varepsilon}$ and not the exponent. Over the same sample the number of a with a nonempty fibre grew as $13, 19, 25, 33$ against a range growing as $38, 64, 161, 330$, consistent with the heuristic $\sum_a d(ap+1)/\varphi(a) \asymp \log^2 p$.

8.1 The count on the modular hyperbola

The decomposition (9) can be run in the other direction, eliminating a instead of u and v. Setting $d = au - 1$ and $e = av - 1$, identity (8) reads $de = ap + 1$, so $de \equiv 1 \bmod p$ and $a = (de-1)/p$, while $a \mid d+1$ and $a \mid e+1$ are exactly the conditions defining u and v.

Proposition 8.7. *$C(p) = \#\big\{(d, e) \in \mathbb{Z}^2_{>0} \,:\, de \equiv 1 \bmod p,\ \frac{de-1}{p} \mid \gcd(d+1, e+1)\big\}$.*

Proof. The maps $(a, u, v) \mapsto (au-1, av-1)$ and $(d, e) \mapsto \big((de-1)/p, (d+1)p/(de-1), (e+1)p/(de-1)\big)$ are mutually inverse by (8). That the second is defined on the whole target set, which is where the divisibilities $(de-1)/p \mid d+1$ and $(de-1)/p \mid e+1$ are used, is checked rather than argued below, so this proposition rests on a computation at that point. Checked against $C(p)$ for all primes $5 \leq p < 120$ with no discrepancy (`code/modular_hyperbola.py`). □

So $C(p)$ counts points of the modular hyperbola $de \equiv 1 \bmod p$ cut by a divisibility that ties the level $a = (de-1)/p$ to $\gcd(d+1, e+1)$. The same substitution turns the sum that every upper bound passes through into a hyperbola count with no side condition,

$$\sum_{a \leq A} d(ap+1) = \#\{(d, e) \in \mathbb{Z}^2_{>0} \,:\, de \equiv 1 \bmod p,\ 1 < de \leq Ap + 1\}, \tag{10}$$

and this locates the wall. Counts of points of $de \equiv 1 \bmod p$ in regions are governed by Kloosterman sums, and Weil's bound gives them with an error $O_\varepsilon(p^{1/2+\varepsilon})$. With $A = (2p)^{1/3}$ the main term of (10) is of size $A \log p$, that is $p^{1/3+o(1)}$. The error term exceeds the main term. Kloosterman technology therefore does not reach the trivial bound $C(p) \ll_\varepsilon p^{1/3+\varepsilon}$ of Theorem 8.3, let alone improve on it, and the improvement would have to come from below the square root barrier rather than from a sharper application of Weil.

This is consistent with the state of the divisor function in arithmetic progressions. Grimmelt and Merikoski [7] reach exponent of distribution $2/3$, but for almost all moduli with two small factors. Conjecture 8.4 needs a bound for one fixed p and every modulus $a \le p^{1/3}$, and averaging over the modulus is exactly what is unavailable.

It is worth being precise about where the exponent $1/3$ is lost, because it is not lost where one might expect. The sum in (10) is genuinely of size $A \log p$: computed at the primes nearest $10^4, 10^5, 10^6, 10^7$ with $A = (2p)^{1/3}$, the values of $\sum_{a \le A} d(ap+1)$ are $348, 906, 2314, 5928$ against $A \log p = 249, 668, 1727, 4368$, a ratio near 1.36 throughout. So Theorem 8.3 does not throw anything away in the summation. Everything is lost in the single step $R_a(p) \le d(ap+1)$, which discards the congruence $D \equiv -1 \bmod a$. The same computation gives $\sum_{a \le A} d(ap+1)$ against $\sum_{a \le A} R_a(p)$ as $4.97, 11.92, 23.61, 47.81$: the ratio doubles with each decade of p, tracking $A = 27, 58, 125, 271$, which is of the order of the average of $\varphi(a)$ over $a \le A$, itself $\sim (3/\pi^2)A$; the measured ratios are about $0.18A$. The whole gap between $p^{1/3}$ and $\log^2 p$ is that one discarded factor (`code/loss_accounting.py`).

This also identifies which technology would have to apply. The one method that goes past Weil's bound in problems of this shape is to average over the modulus and appeal to Kuznetsov's formula, as in Deshouillers and Iwaniec and in Bombieri, Friedlander and Iwaniec. Here the moduli a do vary, over $a \le p^{1/3}$, so averaging is available in principle. What is not available is independence: the number $ap+1$ whose divisors are being counted is itself a function of the modulus a. That linkage, and not the absence of a range of moduli, is what removes the standard route.

Every change of variable applied to $auv = p + u + v$ in the course of this work returns an instance of the same equation. Setting $s = u + v$ and $m = uv$ and splitting the discriminant $(am - p)^2 - 4m$ produces $aXY - 2X - 2Y = 4p$ with $X = 2u$, $Y = 2v$; writing the divisor as $ut - 1$ with cofactor k produces $tuk = p + u + k$; normalising by $g = \gcd(d+1, e+1)$ and writing $d = gD - 1$, $e = gE - 1$ produces $gDE - D - E = p$; and the passage to the modular hyperbola is reversible. Each of these is an algebraic identity and each is formalised (`fixed_point_double`, `fixed_point_reindex`, `fixed_point_gcd`, `fixed_point_hyperbola`). The general assertion, that every elementary reindexing behaves this way, is not a formal statement and is not claimed as one.

The substitution (8) is an identity over any commutative ring (`fixed_point_ring`), which is why the analogue over $\mathbb{F}_q[t]$ inherits the structure of the problem intact rather than simplifying it.

9 Formalization

The development accompanying this paper is a Lean 4 library against Mathlib. It contains no `sorry`, no `native_decide` and no additional axioms: every declaration cited below depends only on `propext`, `Classical.choice` and `Quot.sound`, which continuous integration checks on each commit.

What is machine-checked is the enumeration and the reductions it rests on. In particular the counts are taken over Definition 2.1 itself rather than over a convenient class: the definition is transcribed with no periodicity assumed (periodicity is derived, not assumed, at widths 4 and 5), and the resulting sets are put in bijection with the arithmetic conditions the search enumerates.

Theorem 4.1 and the divisibility statements of Section 6 are verified in that sense.

So is the count at a prime, end to end. Theorem 7.7 and Lemma 8.1 compose in the library into a single declaration giving $T(p,5) = 5 + 5C(p)$ for every prime $p \geq 5$, stated over Definition 2.1 with periodicity derived rather than assumed, and with $C(p)$ the full solution count of $auv = p + u + v$ rather than a count inside a search box.

What is not machine-checked is stated as such. The external inputs of Shiu and of Linnik are cited, not proved; and the passages to O_ε in Theorem 8.3 and Corollary 7.3 are written proofs.

A per-statement table naming, for each result, the Lean declaration that verifies it or recording that none exists, is maintained with the library rather than here, since it changes with the library and not with the paper. Tables and computed constants in this paper are regenerated by the scripts named alongside them.

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